

Data.gov Dataset Database

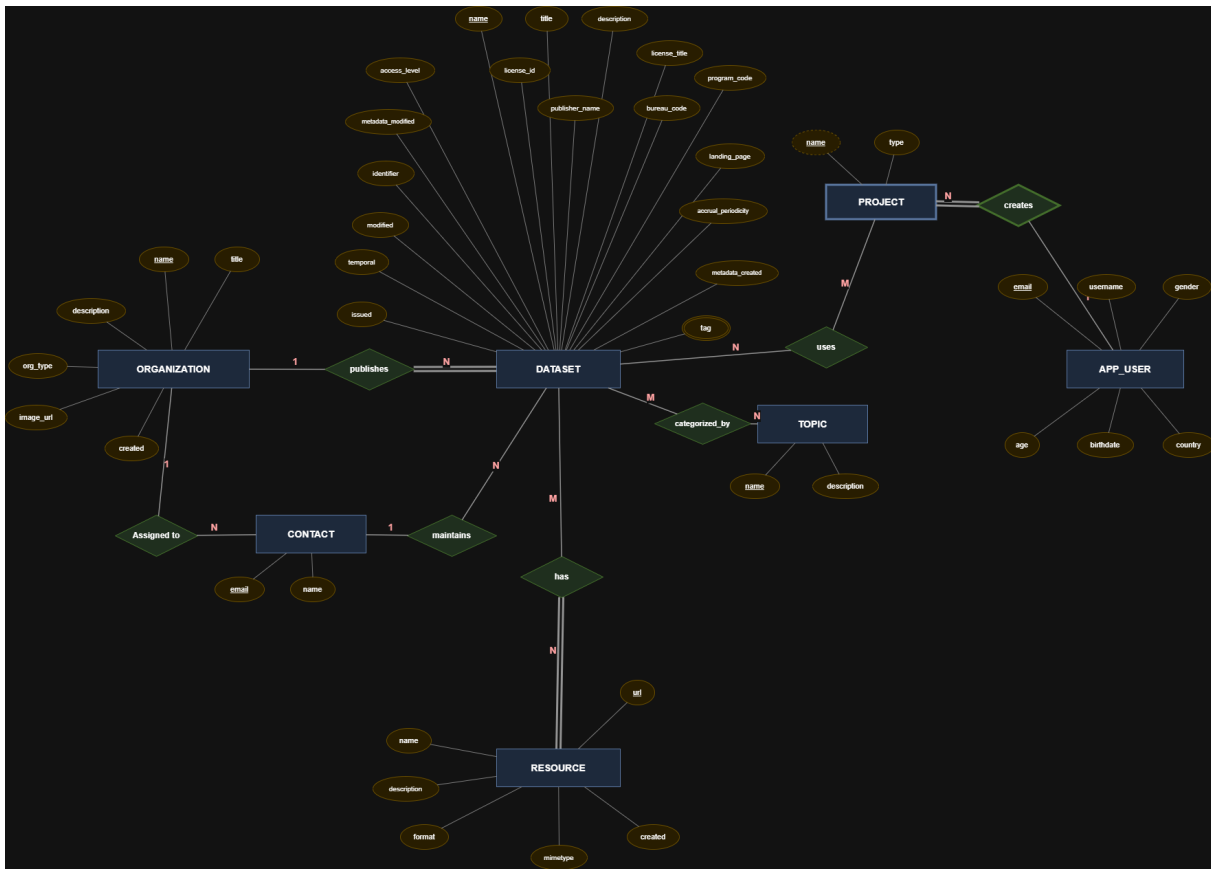
CSCE 2501 : Fundamentals of Database Systems
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1 Entity-Relationship Diagram



1.1 Entities

- **ORGANIZATION** : A US federal agency / publishing body on data.gov. PK: name (slug).
- **CONTACT** : A contact person assigned to an organization who may maintain one or more datasets. PK: email.
- **DATASET** : A dataset (package) published on data.gov. PK: name (slug).
- **RESOURCE** : A downloadable file or API endpoint identified globally by its URL. PK: url.
- **APP_USER** : A registered application user. PK: email.
- **TOPIC** : A topic / theme that datasets can belong to. PK: name (e.g. “Transportation”, “Health”).
- **PROJECT** : A weak entity representing a user’s project. Identified through APP_USER via the *creates* identifying relationship. Partial key: name.

1.2 Relationships

Relationship	Entity 1	Cardinality	Entity 2
publishes	ORGANIZATION	1 : N	DATASET
assigned_to	CONTACT	N : 1	ORGANIZATION
maintains	CONTACT	1 : N	DATASET
has	DATASET	M : N	RESOURCE
categorized_by	DATASET	M : N	TOPIC
uses	PROJECT	M : N	DATASET
creates	APP_USER	1 : N	PROJECT

Table 1: Relationships and their cardinalities

1.3 Multivalued Attributes

- **tag** (of DATASET) : Free-text tags from CKAN.

1.4 Derived Attribute

- **age** (of APP_USER) : Computed from `birthdate`:
`TIMESTAMPDIFF(YEAR, birthdate, CURDATE())`

2 Relational Model

All primary keys are **natural** Underlined attributes form the primary key; *italicised* attributes denote foreign keys.

1. **ORGANIZATION**(name, title, description, org_type, image_url, created)
2. **CONTACT**(email, name, *org_name*)
org_name → ORGANIZATION.name
3. **DATASET**(name, title, description, access_level, license_id, license_title, meta-data_created, metadata_modified, identifier, publisher_name, landing_page, modified, issued, accrual_periodicity, temporal, bureau_code, program_code, *org_name*, *maintainer_email*)
org_name → ORGANIZATION.name
maintainer_email → CONTACT.email
4. **RESOURCE**(url, name, description, format, mimetype, created)
5. **DATASET_RESOURCE**(dataset_name, url) : M:N bridge
dataset_name → DATASET.name
url → RESOURCE.url
6. **DATASET_TAG**(dataset_name, tag_name) : multivalued attribute
dataset_name → DATASET.name
7. **TOPIC**(name, description)
8. **DATASET_TOPIC**(dataset_name, topic_name) : M:N bridge
dataset_name → DATASET.name
topic_name → TOPIC.name
9. **APP_USER**(email, username, gender, birthdate, country)
age = derived from birthdate
10. **PROJECT**(*email*, name, type) : weak entity
email → APP_USER.email
11. **PROJECT_DATASET**(*email*, project_name, dataset_name) : M:N “uses”
(email, project_name) → PROJECT.(email, name)
dataset_name → DATASET.name

3 Database Schema (MySQL DDL)

The complete DDL is stored in `schema.sql` and reproduced below.

```
1  -- =====
2  -- Data.gov Dataset Database Schema
3  -- Database: datagov_db
4  -- MySQL DDL Statements
5  -- Course: CSCE 2501      Fundamentals of Database Systems
6  -- Student: Mahmoud Alaskndrani (900241828)
7  -- =====
8
9  DROP DATABASE IF EXISTS datagov_db;
10 CREATE DATABASE datagov_db CHARACTER SET utf8mb4 COLLATE
    utf8mb4_unicode_ci;
11 USE datagov_db;
12
13 -- =====
14 -- 1. ORGANIZATION (Entity)
15 -- Represents a US federal agency / publishing body on data.gov.
16 -- PK: name      the unique organization slug (e.g. "epa-gov").
17 -- =====
18 CREATE TABLE organization (
19     name          VARCHAR(255)      NOT NULL,
20     title          VARCHAR(500),
21     description     TEXT,
22     org_type        VARCHAR(100),
23     image_url       VARCHAR(500),
24     created         DATETIME,
25     PRIMARY KEY (name)
26 );
27
28 -- =====
29 -- 2. CONTACT (Entity)
30 -- A contact person assigned to an organization who may
31 -- maintain one or more datasets.
32 -- PK: email.  FK: org_name -> ORGANIZATION (assigned_to).
33 -- =====
34 CREATE TABLE contact (
35     email          VARCHAR(255)      NOT NULL,
36     name           VARCHAR(255),
37     org_name        VARCHAR(255),
38     PRIMARY KEY (email),
39     CONSTRAINT fk_contact_org
40         FOREIGN KEY (org_name) REFERENCES organization(name)
41         ON DELETE SET NULL ON UPDATE CASCADE
42 );
43
44 -- =====
45 -- 3. DATASET (Entity)
46 -- Core dataset (package) information crawled from data.gov.
47 -- PK: name      the unique dataset slug.
48 -- FK org_name -> ORGANIZATION (publishes).
49 -- FK maintainer_email -> CONTACT (maintains).
50 -- =====
51 CREATE TABLE dataset (
52     name            VARCHAR(255)      NOT NULL,
53     title            VARCHAR(500),
```

```

54     description          TEXT,
55     access_level         VARCHAR(50),
56     license_id           VARCHAR(100),
57     license_title        VARCHAR(255),
58     metadata_created     DATETIME,
59     metadata_modified    DATETIME,
60     identifier            VARCHAR(500),
61     publisher_name        VARCHAR(500),
62     landing_page          VARCHAR(1000),
63     modified              DATE,
64     issued                DATE,
65     accrual_periodicity   VARCHAR(50),
66     temporal              VARCHAR(255),
67     bureau_code           VARCHAR(50),
68     program_code          VARCHAR(50),
69     org_name              VARCHAR(255),
70     maintainer_email      VARCHAR(255),
71     PRIMARY KEY (name),
72     CONSTRAINT fk_dataset_org
73         FOREIGN KEY (org_name) REFERENCES organization(name)
74         ON DELETE SET NULL ON UPDATE CASCADE,
75     CONSTRAINT fk_dataset_maintainer
76         FOREIGN KEY (maintainer_email) REFERENCES contact(email)
77         ON DELETE SET NULL ON UPDATE CASCADE
78 );
79
80 -- =====
81 -- 4. RESOURCE (Entity)
82 -- Downloadable files / API endpoints identified by URL.
83 -- PK: url (natural key).
84 -- =====
85 CREATE TABLE resource (
86     url          VARCHAR(500) NOT NULL,
87     name          VARCHAR(500),
88     description    TEXT,
89     format         VARCHAR(100),
90     mimetype       VARCHAR(100),
91     created        DATETIME,
92     PRIMARY KEY (url)
93 );
94
95 -- =====
96 -- 5. DATASET_RESOURCE (M:N relationship DATASET <-> RESOURCE)
97 -- Links datasets to resources.
98 -- Composite PK: (dataset_name, url).
99 -- =====
100 CREATE TABLE dataset_resource (
101     dataset_name    VARCHAR(255) NOT NULL,
102     url             VARCHAR(500) NOT NULL,
103     PRIMARY KEY (dataset_name, url),
104     CONSTRAINT fk_datasetresource_dataset
105         FOREIGN KEY (dataset_name) REFERENCES dataset(name)
106         ON DELETE CASCADE ON UPDATE CASCADE,
107     CONSTRAINT fk_datasetresource_resource
108         FOREIGN KEY (url) REFERENCES resource(url)
109         ON DELETE CASCADE ON UPDATE CASCADE
110 );
111

```

```

112 -- =====
113 -- 6. DATASET_TAG (Multivalued attribute "tag" of DATASET)
114 -- Each dataset may carry multiple free-text tags from CKAN.
115 -- Composite PK: (dataset_name, tag_name).
116 -- =====
117 CREATE TABLE dataset_tag (
118     dataset_name    VARCHAR(255)    NOT NULL,
119     tag_name        VARCHAR(255)    NOT NULL,
120     PRIMARY KEY (dataset_name, tag_name),
121     CONSTRAINT fk_datasettag_dataset
122         FOREIGN KEY (dataset_name) REFERENCES dataset(name)
123         ON DELETE CASCADE ON UPDATE CASCADE
124 );
125
126 -- =====
127 -- 7. TOPIC (Entity)
128 -- A topic / theme that datasets can belong to.
129 -- PK: name the topic label (e.g. "Transportation", "Health").
130 -- =====
131 CREATE TABLE topic (
132     name            VARCHAR(255)    NOT NULL,
133     description     TEXT,
134     PRIMARY KEY (name)
135 );
136
137 -- =====
138 -- 8. DATASET_TOPIC (M:N relationship DATASET <-> TOPIC)
139 -- Links datasets to topics.
140 -- Composite PK: (dataset_name, topic_name).
141 -- =====
142 CREATE TABLE dataset_topic (
143     dataset_name    VARCHAR(255)    NOT NULL,
144     topic_name      VARCHAR(255)    NOT NULL,
145     PRIMARY KEY (dataset_name, topic_name),
146     CONSTRAINT fk_datasettopic_dataset
147         FOREIGN KEY (dataset_name) REFERENCES dataset(name)
148         ON DELETE CASCADE ON UPDATE CASCADE,
149     CONSTRAINT fk_datasettopic_topic
150         FOREIGN KEY (topic_name) REFERENCES topic(name)
151         ON DELETE CASCADE ON UPDATE CASCADE
152 );
153
154 -- =====
155 -- 9. APP_USER (Entity)
156 -- Registered application users.
157 -- PK: email. age is a derived attribute (from birthdate).
158 -- =====
159 CREATE TABLE app_user (
160     email           VARCHAR(255)    NOT NULL,
161     username        VARCHAR(100)    NOT NULL,
162     gender          ENUM('Male', 'Female') NOT NULL,
163     birthdate       DATE,
164     country         VARCHAR(100),
165     PRIMARY KEY (email),
166     UNIQUE KEY uk_user_username (username)
167 );
168
169 -- =====

```

```

170 -- 10. PROJECT (Weak Entity identified through APP_USER)
171 -- A user's project. Partial key: project_name.
172 -- Full PK: (email, project_name).
173 -- FK email -> APP_USER (creates identifying relationship).
174 -- =====
175 CREATE TABLE project (
176     email          VARCHAR(255)      NOT NULL,
177     project_name    VARCHAR(255)      NOT NULL,
178     project_type    ENUM('analytics', 'machine_learning', '
179         field_research') NOT NULL,
180     PRIMARY KEY (email, project_name),
181     CONSTRAINT fk_project_user
182         FOREIGN KEY (email) REFERENCES app_user(email)
183         ON DELETE CASCADE ON UPDATE CASCADE
184 );
185 -- =====
186 -- 11. PROJECT_DATASET (M:N relationship "uses": PROJECT <-> DATASET)
187 -- Links projects to datasets they use.
188 -- Composite PK: (email, project_name, dataset_name).
189 -- =====
190 CREATE TABLE project_dataset (
191     email          VARCHAR(255)      NOT NULL,
192     project_name    VARCHAR(255)      NOT NULL,
193     dataset_name    VARCHAR(255)      NOT NULL,
194     PRIMARY KEY (email, project_name, dataset_name),
195     CONSTRAINT fk_projds_project
196         FOREIGN KEY (email, project_name) REFERENCES project(email,
197             project_name)
198         ON DELETE CASCADE ON UPDATE CASCADE,
199     CONSTRAINT fk_projds_dataset
200         FOREIGN KEY (dataset_name) REFERENCES dataset(name)
201         ON DELETE CASCADE ON UPDATE CASCADE,
202     INDEX idx_projds_dataset (dataset_name)
203 );

```